

NIKUL PATEL

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EDUCATION

Clarkson University

Double Major: Aerospace and Mechanical Engineering | Minor: Robotics | Honors Program
Clubs and Organizations: Society of Asian Scientists and Engineers | Underwater Robotics

Potsdam, NY

Expected Graduation: 2028

GPA: 3.96

PROFESSIONAL EXPERIENCE

Co-Op Student

Baldwin Richardson Foods Co.

Jan 2026 – Present

Williamson, NY

- Leading a DMAIC-based initiative to standardize Lockout/Tagout (LOTO) procedures by identifying gaps, implementing corrective actions, training personnel, and ensuring long-term compliance and sustainability.
- Contributing to cross-functional Six Sigma-driven continuous improvement projects, participating in Kaizen events to reduce automated asset downtime, conducting Root Cause Analysis (RCA), and implementing corrective actions to improve reliability and operational efficiency.

Design Build Captain

Engineering Design Team

Aug 2025 – Present

Potsdam, NY

- Designed and 3D-printed custom mounts and fixtures for cameras, IMUs, pressure sensors, and control boards using SolidWorks and OnShape to ensure reliability in underwater environments.
- Worked closely with electrical and controls teams to integrate wiring, connectors, and microcontrollers into mechanical layouts for clean, serviceable robot builds.

Undergraduate Research Assistant

Photoacoustic Research Laboratory

Jan 2025 – May 2025

Potsdam, NY

- Designed and 3D-printed mechanical fixtures to mount and align ultrasonic sensors for automated data collection, improving measurement stability and repeatability by 20%.
- Worked with controls engineering students to integrate hardware with PLC systems and ran iterative system tests to improve reliability, throughput, and scalability.

Research Intern

Spartificial

Mar 2024 – July 2024

Remote

- Analyzed and organized a dataset of 22,000+ samples across 22 classes and 4 timeframes; independently designed a Late Fusion multi-view CNN while leading a 2-member team on Early & Intermediate Fusion models.
- Achieved 95%+ accuracy for each model and 20% faster training, while optimization experiments improved predictive accuracy by 10% and processing efficiency by 15%.

Aerospace Structures Design Intern

Feynman Aerospace

Oct 2023 – Dec 2023

Remote

- Designed and assembled a 1U CubeSat structural model in SolidWorks, incorporating deployable antennas, solar panels, and frame assemblies; conducted comparative material evaluation (Al 7075-T6, CFRP, Ti alloys) to balance strength-to-weight and manufacturability.
- Performed finite element static and thermal load analyses in Ansys under launch and orbital worst-case scenarios, demonstrating a possibility of 25% increase in structural reliability and a 15% reduction in material usage.

LEADERSHIP AND ACADEMIC ENGAGEMENT

Teaching Assistant | MATLAB (HP102/103) | Clarkson University

Aug 2025 – Present

- Co-Managing a fully TA-led honors course | Coordinating lectures, curriculum, assignments, grading, and office hours.

Class Senator | Student Association | Clarkson University

Sept 2025 – Present

- Advocating for student interests through active participation in policy discussions and voting on key campus issues.

SKILLS

Engineering: CAD (SolidWorks, Onshape), FEA, Structural & Thermal Simulation, DFM, Rapid Prototyping

Robotics & Programming: Sensor Integration, Embedded Systems (Arduino), MATLAB/Simulink, Python

Process & Operations: Lean/Six Sigma (DMAIC, RCA/RCCA), Engineering Data Analysis, Process Optimization, Kaizen

AWARDS

Presidential Scholar: Academic Excellence | *Clarkson University*

All Semesters

SASE Rising Leader: Recognition for strong leadership skills and proactive initiative | *Clarkson University*

April 2025

Master BSMS: Nominated Best student of outgoing batch in all aspects. | *B.S. Memorial School*

Feb 2023